

Synthesis of Aluminum Metal-Organic Frameworks

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ABSTRACT. This sample article is provided for text-mining demonstrations. MOF-303 and CAU-23 illustrate the preparation of aluminum metal-organic frameworks from small vessels to larger reactors. All quantities, conditions, and outcomes are for demonstration purposes. Reagent choices and synthesis descriptions draw on existing literature.

INTRODUCTION

Water adsorption in metal-organic frameworks supports atmospheric water harvesting.^{1,2} Demonstrations powered by sunlight³ and systems with rapid cycling⁴ connect adsorption behavior with practical water collection and motivate scalable preparation.

Mixed-component frameworks allow linker composition to adjust pore chemistry and hydrophilicity.^{5,6} Studies of aluminum frameworks also identify limits to linker substitution⁷ and trace the development of crystalline order.⁸

MOF-303 and CAU-23 connect aluminum-containing chains with organic linkers to form porous solids. Their pore environments reflect the linker chemistry and chain arrangement.⁹

This sample writing is intended for text-mining demonstrations. Detailed preparations appear in the Methods section; characterization details are provided in the Supporting Information.

RESULTS AND DISCUSSION

MOF-303 is constructed from aluminum centers and pyrazole-3,5-dicarboxylate linkers (Figure 1a).^{10,11} CAU-23 contains thiophene-2,5-dicarboxylate in the corresponding organic component (Figure 1b).¹² In both frameworks, connected AlO_6 octahedra form extended inorganic chains that are joined by the organic linkers.

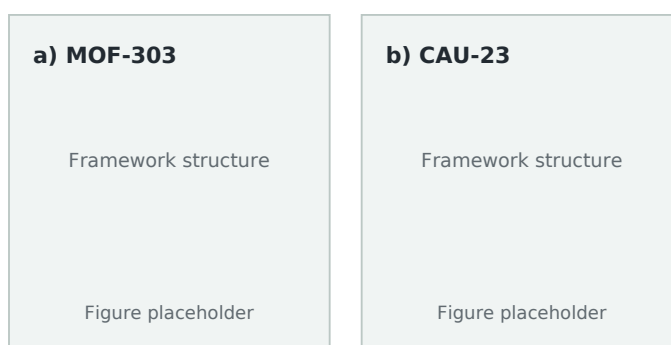


Figure 1. Framework structures of (a) MOF-303 and (b) CAU-23. Boxes reserve space for the structural views.

The chains differ in the arrangement of their shared corners. MOF-303 alternates between cis and trans connections, whereas CAU-23 follows a sequence of four cis and four trans connections. The pyrazole nitrogen atoms and thiophene sulfur atom also give the two pore environments different chemical character.

Despite their structural differences, both frameworks can be prepared using aqueous aluminum chloride and the corresponding linker in sodium hydroxide solution.⁹ At the vial scale, 0.5 mmol of linker and 0.5 mmol of aluminum chloride were heated at 120 °C for 12 hours. Washing and activation afforded 49 mg of MOF-303 at 49% yield and 53 mg of CAU-23 at 50% yield (Table 1).

Sodium hydroxide served as the base and pH modulator by promoting linker deprotonation before metal addition. The vial preparations used 1.5 equivalents per linker, while the flask and reactor routes used 3 equivalents. The higher base loading accounts for linker deprotonation and the hydroxide groups in the aluminum chains.⁹

RESULTS AND DISCUSSION (CONTINUED)

The development of crystalline order depended on the duration of heating. No crystalline materials formed when the reaction time was less than 1 hour. The appearance of a precipitate alone thus did not establish framework formation.

For gram-scale synthesis, aluminum solution was added slowly over 2 h to 10 mmol of linker, followed by 3 h of reflux.^{9,13} After workup, MOF-303 gave 1.87 g at 94% yield and CAU-23 gave 1.93 g at 90% yield (Table 1; Methods).

The same addition sequence was extended to 20 mol of linker in a 200 L reactor. Reflux for 6 h and subsequent workup afforded 3.59 kg of MOF-303 at 91% yield and 3.61 kg of CAU-23 at 84% yield (Table 1).⁹

PXRD assessed crystalline order and phase identity (Figure 2a), while nitrogen sorption at 77 K examined accessible porosity (Figure 2b). Additional characterization included TGA and SEM (Figures S3 and S4).

It was also observed that products prepared at reaction temperatures below 100 °C had no crystallinity, as confirmed by PXRD (Figure S1). For scale-up, the 120 °C heating setting used in the small-scale flask preparations was retained. In the reactor, this was the jacket temperature; the reaction mixture reached 100 °C.

After reflux, the reactor mixture was cooled to 60 °C and filtered. Washing with aqueous ethanol was followed by dispersion in anhydrous ethanol and overnight drying in air. The powder was then heated at 120 °C for 48 h. The flask preparations used water and ethanol washing, followed by activation at 120 °C for 24 h under dynamic vacuum. Detailed synthesis and isolation procedures are given in Methods; characterization procedures are provided in Sections S1-S4 of the Supporting Information.

Table 1. Isolated masses and yields of MOF-303 and CAU-23 at three preparation scales.

Framework	Preparation scale	Isolated mass	Yield (%)
MOF-303	Vial	49 mg	49
CAU-23	Vial	53 mg	50
MOF-303	Flask	1.87 g	94
CAU-23	Flask	1.93 g	90
MOF-303	Reactor	3.59 kg	91
CAU-23	Reactor	3.61 kg	84



Figure 2. Characterization of MOF-303 and CAU-23 by (a) powder X-ray diffraction and (b) nitrogen sorption at 77 K. The panels are placeholders and contain no measured data.

METHODS

Reagents

MOF-303 used 1H-pyrazole-3,5-dicarboxylic acid monohydrate ($\text{H}_2\text{PZDC}\cdot\text{H}_2\text{O}$), and CAU-23 used thiophene-2,5-dicarboxylic acid (H_2TDC). The laboratory linker reagents had listed purities of at least 98%; aluminum chloride hexahydrate and sodium hydroxide had listed purities of at least 99% and 97%, respectively. Water was the reaction solvent, and NaOH supplied the base and pH modulator.

Vial synthesis

For each framework, a 20 mL vial was charged with 0.5 mmol of its corresponding linker: $\text{H}_2\text{PZDC}\cdot\text{H}_2\text{O}$ for MOF-303 or H_2TDC for CAU-23. The linker was dissolved in 9.5 mL of 0.079 M aqueous NaOH, corresponding to 1.5 base equivalents. Addition of 0.5 mL of 1 M aqueous aluminum chloride supplied 0.5 mmol of AlCl_3 . The clear mixture was heated in a preheated oven at 120 °C for 12 hours.

The solid was washed with three 15 mL portions of water over one day and three 15 mL portions of methanol over another day. Under dynamic vacuum (approximately 10^{-3} mbar), the temperature was raised to 120 °C over 6 h. MOF-303 gave 49 mg at 49% yield, and CAU-23 gave 53 mg at 50% yield.

Flask reflux synthesis

A 250 mL round-bottom flask contained 10 mmol of the corresponding linker dissolved in 50 mL of 0.6 M aqueous NaOH (3 base equivalents). A solution of $\text{AlCl}_3\cdot 6\text{H}_2\text{O}$ (2.41 g, 10 mmol) in 50 mL water was added dropwise over 2 h at room temperature under vigorous stirring. A white precipitate appeared. The heating setting was 120 °C, and reflux continued for 3 h.

After cooling to room temperature, the flask products were collected by centrifugation. Each solid was washed twice with water and three times with ethanol, then dried in air overnight. Activation at 120 °C for 24 h under dynamic vacuum (approximately 10^{-3} mbar) gave 1.87 g of MOF-303 at 94% yield or 1.93 g of CAU-23 at 90% yield. The flask liquid temperature was not separately specified.

Reactor synthesis

A 200 L glass reactor was charged with 20 mol of H_2PZDC for MOF-303 or 20 mol of H_2TDC for CAU-23. NaOH (2.4 kg, 60 mol; 3 base equivalents) and 38 L water were added. Stirring for 60 min completed dissolution and allowed the mixture to return to room temperature.

A feed of $\text{AlCl}_3\cdot 6\text{H}_2\text{O}$ (4.82 kg, 20 mol) dissolved in 12 L water was introduced through a 15 L glass feed funnel at 6 L per hour over 2 h. The stirring speed was 100 rpm, and a white precipitate formed. The jacket was set to 120 °C; the reaction mixture reached 100 °C and was refluxed for 6 h.

The mixture was cooled to 60 °C and filtered through a 20 L filtration funnel. The powder was washed with 15 L of 70% ethanol in water by volume, then dispersed in 30 L of anhydrous ethanol and stirred at room temperature. After filtration and overnight drying in air, the material was heated in an oven at 120 °C for 48 h. MOF-303 gave 3.59 kg at 91% yield, and CAU-23 gave 3.61 kg at 84% yield.

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